

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Fall 2023
Computer Organization & Assembly Language (CS-215T)
Assignment # 3

Semester: 3rd
Announced Date: 19-01-2024
Total Marks: 10
Instructor Name: Shakir Karim / Dr. Abdul Hameed Pitafi

Batch: 2022F
Due Date: 26-01-2024
Marks Obtained:

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom’s Taxonomy
CLO 3	Apply the elements and design concepts of cache memory using different mapping techniques to improve the performance of computer system.	PLO_3 (Problem Analysis)	C3 (Applying)

Question 1 **[10 marks]**
Consider the size of main memory as 128 Bytes and the size of cache memory as 32 Bytes with line size of 4 bytes. Assuming the Set-Associative mapped cache, answer the following:

- a) (1 mark)
Write the formula to map block of main memory to set of cache memory
- b) (4 marks)
Calculate the block size, line size, set size, tag size
- c) (4 marks)
Show the distribution of cache memory in desired sets containing tag numbers, respective blocks, and byte numbers with the help of diagram
- d) (1 mark)
Show the tag allocation for each set of cache memory